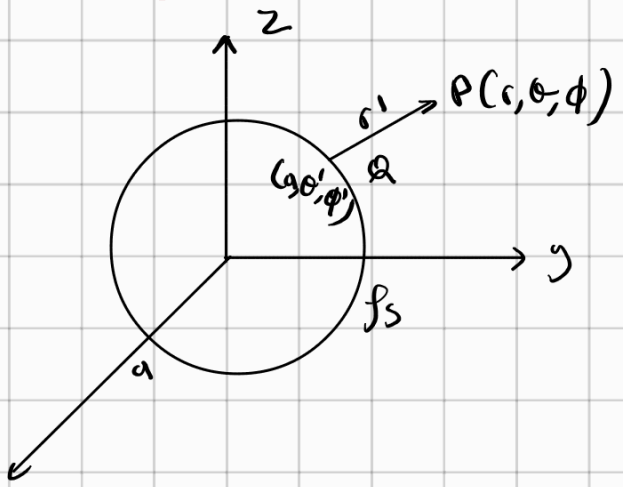
S.27 Pr.10 ~~A~~ (4.65) p.2.3-20



a) $V = ?$

$$Q = \int_{\theta=0}^{\pi}\int_{\phi=0}^{2\pi} \rho_s \, ds$$

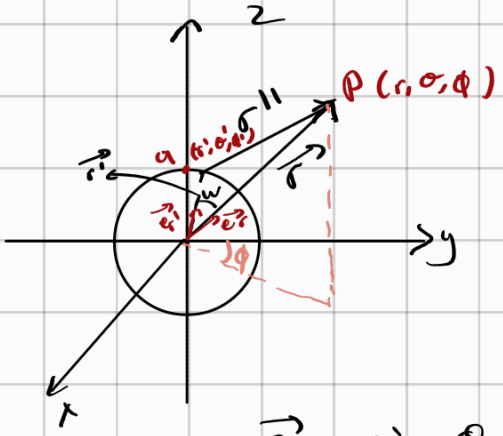
$$ds = a^2 \sin\theta \, d\phi \, d\theta$$

$$= \int_{\theta=0}^{\pi}\int_{\phi=0}^{2\pi} \rho_s a^2 \sin\theta \, d\phi \, d\theta$$

$$\Rightarrow Q = 4\pi a^2 \rho_s \Rightarrow \rho_s = \frac{Q}{4\pi a^2}$$

$$dV = \frac{1}{4\pi\varepsilon_0}\frac{dQ}{r'} = \frac{1}{4\pi\varepsilon_0}\frac{\rho_s \, ds}{r'}$$

$$V = \int_{\theta=0}^{\pi}\int_{\phi=0}^{2\pi} dV = \int_{\theta=0}^{\pi}\int_{\phi=0}^{2\pi} \frac{\rho_s}{4\pi\varepsilon_0} a^2 \sin\theta' \, d\theta' \, d\phi'$$



$$r' = \sqrt{a^2 + r^2 - 2a\cos\psi}$$

$$= r\sqrt{1 + \frac{a^2}{r^2} - \frac{2a\cos\psi}{r}}$$

$\cos\psi$

$$\vec{e}_r \cdot \vec{e}_{r'} = |\vec{e}_r| \cdot |\vec{e}_{r'}| \cos\psi = \cos\psi$$

$$\vec{e}_r = \sin\theta\cos\phi\,\vec{e}_x + \sin\theta\sin\phi\,\vec{e}_y + \cos\theta\,\vec{e}_z$$

$$\vec{e}_{r'} = \sin\theta'\cos\phi'\,\vec{e}_z + \sin\theta'\sin\phi'\,\vec{e}_y + \cos\theta'\,\vec{e}_z$$

$$\vec{e}_r \cdot \vec{e}_{r'} = \sin\theta\sin\theta'\cos\phi\cos\phi' + \sin\theta\sin\theta'\sin\phi\sin\phi' + \cos\theta\cos\theta'$$

$$\cos\psi = \sin\theta\sin\theta'\cos(\phi' - \phi) + \cos\theta\cos\theta'$$

Basitlik olması için P noktası z ekseni üzerinde olsun. $\theta = 0$ olur

$$\Rightarrow \cos\psi = \cos\theta'$$

$$V = \frac{Q}{16\pi^2\varepsilon_0} \frac{2\pi}{\varepsilon_0} \int_{\theta'=0}^{\pi} \frac{\sin\theta' \, d\theta'}{\sqrt{a^2+r^2-2ar\cos\theta'}} \, d\theta'$$

$$V = \frac{Q}{16\pi^2\varepsilon_0} \frac{2\pi}{\varepsilon_0} \frac{1}{2ar} \int_{\theta'=0}^{\pi} \frac{2ar\sin\theta' \, d\theta'}{\sqrt{a^2+r^2-2ar\cos\theta'}} \, d\theta'$$

$$= \frac{Q}{8\pi\varepsilon_0 ar} \left(\frac{\sqrt{a^2+r^2+2ar}}{\sqrt{(a+r)^2}} - \frac{\sqrt{a^2+r^2-2ar}}{\sqrt{(a-r)^2}} \right)$$

$$V = \frac{Q}{8\pi\varepsilon_0 ar} \left(|a+r| - |a-r| \right)$$ z ekseni üzerinde

b) $\vec{E}$? $\quad r > a \quad V = \frac{Q}{4\pi\varepsilon_0 r}$

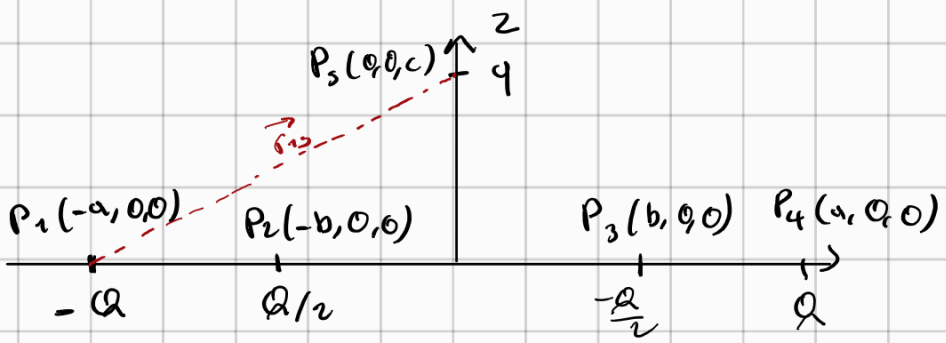
$$\vec{E} = -\text{grad}\, V = -\frac{\partial V}{\partial r} \vec{e}_r$$

$$\boxed{\vec{E} = \frac{Q}{4\pi\varepsilon_0 r^2} \vec{e}_r}$$

c) V ve $\vec{E}$ nin r'ye göre değişimi

$$V = \begin{cases} \frac{Q}{4\pi\varepsilon_0 r}, & r > a \\ \frac{Q}{4\pi\varepsilon_0 a}, & r < a \end{cases} \qquad \vec{E} = \begin{cases} \frac{Q}{4\pi\varepsilon_0 r^2} \vec{e}_r, & \\ 0, & \end{cases}$$

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a) q yükünc ethieden $\vec{F}$ nedir?

$$\vec{F}_{15} = \frac{1}{4\pi\varepsilon_0}\frac{(-qQ)}{r_{15}^3}\vec{r}_{15}$$

$$\vec{r}_{15} = (0-(-a))\vec{e}_x + (c-0)\vec{e}_z = a\vec{e}_x + c\vec{e}_z$$

$$r_{15} = \sqrt{a^2+c^2}$$

$$\boxed{\vec{F}_{15} = \frac{Qq}{4\pi\varepsilon_0}\frac{1}{(a^2+c^2)^{3/2}}(-a\vec{e}_x - c\vec{e}_z)}$$

$\vec{r}_{25} = \vec{P_5} - \vec{P_2}$

$$\vec{r}_{25} = b\vec{e}_x + c\vec{e}_z, \quad r_{25} = \sqrt{b^2+c^2}$$

$$\boxed{\vec{F}_{25} = \frac{Qq}{4\pi\varepsilon_0}\frac{1}{(b^2+c^2)^{3/2}}(b\vec{e}_x + c\vec{e}_z)}$$

$$\vec{r}_{35} = -b\vec{e}_x + c\vec{e}_z, \quad r_{35} = \sqrt{b^2+c^2}$$

$$\vec{F}_{35} = \frac{Qq}{4\pi\varepsilon_0}\frac{1}{(b^2+c^2)^{3/2}}\left(+\frac{b}{2}\vec{e}_x - \frac{c}{2}\vec{e}_z\right)$$

$$\vec{r}_{45} = -a\vec{e}_x + c\vec{e}_z, \quad r_{45} = \sqrt{a^2+c^2}$$

$$\vec{F}_{45} = \frac{Qq}{4\pi\varepsilon_0}\frac{1}{(a^2+c^2)^{3/2}}(-a\vec{e}_x + c\vec{e}_z)$$

$$\vec{F} = \vec{F}_{15} + \vec{F}_{25} + \vec{F}_{35} + \vec{F}_{45}$$

$$\vec{F} = \frac{Qq}{4\pi\varepsilon_0}\left[\frac{1}{(a^2+c^2)^{3/2}}(-a\vec{e}_x - c\vec{e}_z)\right.$$

$$+ \frac{1}{(b^2+c^2)^{3/2}}\left(\frac{1}{2}b\vec{e}_x + \frac{1}{2}c\vec{e}_z\right)$$

$$\left. + \frac{1}{(b^2+c^2)^{3/2}}\left(\frac{1}{2}b\vec{e}_x - \frac{1}{2}c\vec{e}_z\right)\right.$$

$$+\frac{1}{(a^2+c^2)^{3/2}}(-a\vec{e}_x + c\vec{e}_z)\Big]$$

$$\boxed{\vec{F} = \frac{Qq}{4\pi\varepsilon_0}\left(\frac{-2a}{(a^2+c^2)^{3/2}} + \frac{b}{(b^2+c^2)^{3/2}}\right)}$$

b) q yükünün potansiyel enerjisi nedir?

$$V_1 = \frac{-Q}{4\pi\varepsilon_0}\frac{1}{\sqrt{a^2+c^2}} \qquad V_3 = \frac{-Q}{4\pi\varepsilon_0}\frac{1}{2\sqrt{b^2+c^2}}$$

$$V_2 = \frac{Q}{4\pi\varepsilon_0}\frac{1}{2\sqrt{b^2+c^2}} \qquad V_4 = \frac{Q}{4\pi\varepsilon_0}\frac{1}{\sqrt{a^2+c^2}}$$

$$V_T = 0$$

c) $a = 4b$ halinde Oy ekseni üzerinde hangi noktada hangi noktada $\vec{F}=0$ olur? $P_3(0,y,0)$

$$\vec{F} = \frac{Qq}{4\pi\varepsilon_0}\left(\frac{-2a}{(a^2+y^2)^{3/2}} + \frac{b}{(b^2+y^2)^{3/2}}\right)\vec{e}_x$$

$$= \frac{Qq}{4\pi\varepsilon_0}\left(\frac{-8b}{(16b^2+y^2)^{3/2}} + \frac{b}{(b^2+y^2)^{3/2}}\right)\vec{e}_x = 0$$

$$= \frac{8b}{(16b^2+y^2)^{3/2}} = \frac{b}{(b^2+y^2)^{3/2}}$$

$$8(b^2+y^2)^{3/2} = (16b^2+y^2)^{3/2}$$

$$64(b^2+y^2)^3 = (16b^2+y^2)^3$$

$$4(b^2+y^2) = 16b^2+y^2$$

$$\Rightarrow 3y^2 = 12b^2$$

$$y^2 = 4b^2$$

$$\boxed{y = \pm 2b}$$ noktalarında $\vec{F}=0$ olur

Sayfa 32. 1, 2, 3 sorularını çözeceğiz

Gauss

$$\int_S \vec{D} \cdot d\vec{s} = Q \qquad \vec{D} = \varepsilon_0 \vec{E}$$



$$Q = \int_V \rho_v \, dV$$

$$\text{div}\,\vec{D} = \rho$$

$\vec{D}$: deplansman vektörü

ε_0 : Boş uzayın elektriksel geçirgenliği (dielectric permettivity)

$\Delta V = 0$ laplacian denklemi

kapalı yüzeyin içinde yük yok

$\Delta V = -\frac{\rho}{\varepsilon_0}$: poisson denklemi